

THE KING'S SCHOOL

2007
Higher School Certificate
Trial Examination

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question
- Answer each question in a separate booklet

Total marks – 120

- Attempt Questions 1-8
- All questions are of equal value

Blank Page

Total marks – 120

Attempt Questions 1-8

All questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Question 1 (15 marks) Use a SEPARATE writing booklet.

Marks

(a) Use the substitution $t = \tan \frac{x}{2}$ to evaluate $\int_0^{\frac{\pi}{2}} \frac{dx}{3 + \cos x}$ **3**

☐ (b) (i) Evaluate $\int_0^{\frac{\sqrt{3}}{2}} \frac{x}{\sqrt{1-x^2}} dx$ **3**

(ii) Evaluate $\int_0^{\frac{\sqrt{3}}{2}} \cos^{-1} x \, dx$ **2**

(c) (i) Find A if $\frac{2x + A}{x^2 + 1} - \frac{2}{x - 2} \equiv \frac{x - 12}{(x^2 + 1)(x - 2)}$ **1**

☐ (ii) Evaluate $\int_0^1 \frac{x - 12}{(x^2 + 1)(x - 2)} dx$ **3**

(d) Find $\int \tan^3 x \, dx$ **3**

End of Question 1

(a) Let $z = a + i$ and $w = 1 + ai$, a real. Find

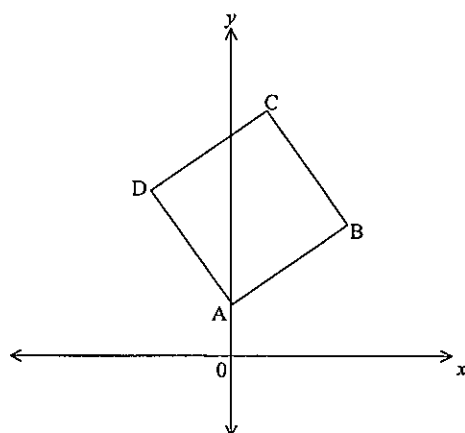
(i) $\left| \frac{z}{w} \right|$ 1

(ii) $\arg zw$ 2

(b) The point $P(x, y)$ represents the complex number z in the Argand diagram.

Sketch the locus of P if $\operatorname{Im}(1 - i)z \geq 1$ 3

(c)



The diagram shows the square ABCD in the complex plane.

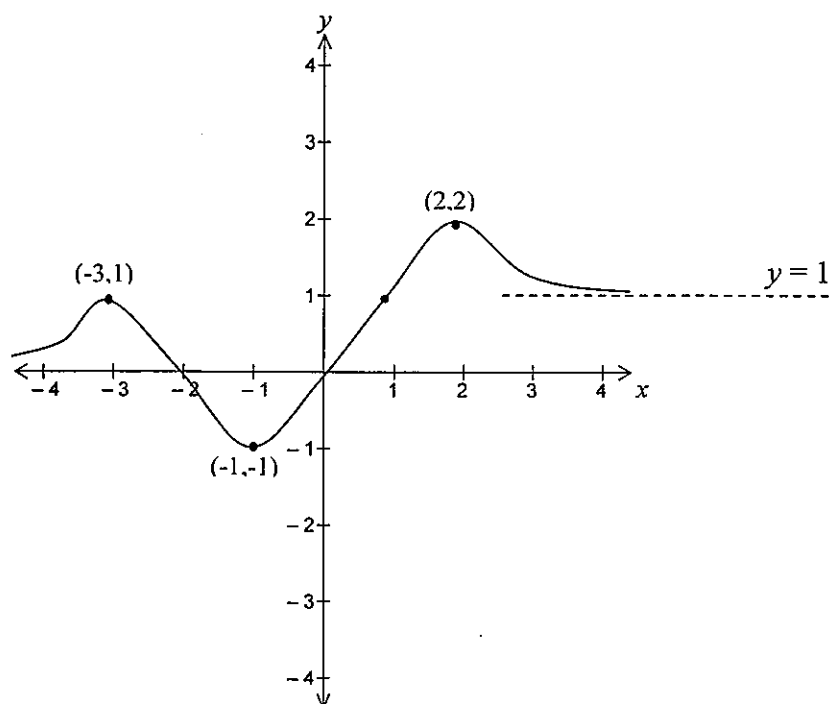
A represents the complex number i and B represents the complex number z .

(i) Find the complex number $\overrightarrow{AD}$ 2

(ii) Hence or otherwise find the complex number represented by the point C. 1

Question 2 continues next page

(d) The diagram shows the graph of $y = f(x)$



The lines $y = 1$ and the x axis are asymptotes.

Draw separate sketches of the graphs of:

(i) $y = \frac{1}{f(x)}$ 2

(ii) $y = \ln f(x)$ 2

(iii) $y = f(x + |x|)$ 2

End of Question 2

- (a) (i) $S(c, 0)$ and $S^1(-c, 0)$, where $c > 0$, are the foci of the hyperbola $x^2 - y^2 = 2$

Find S and sketch the hyperbola showing its foci, directrices, asymptotes and any intercepts made with the coordinate axes.

3

- (ii) $P(\sqrt{2} \sec \theta, \sqrt{2} \tan \theta)$ is a point in the first quadrant on the hyperbola $x^2 - y^2 = 2$ in (i).

A circle with centre P and radius PS is drawn.

- (α) Find the length of the radius in simplest form.

2

- (β) The line S^1P cuts the circle at Q and R where Q is between S^1 and P .

It can be shown that $Q = \left(\frac{2}{\sqrt{2} \sec \theta + 1}, \frac{2\sqrt{2} \tan \theta}{\sqrt{2} \sec \theta + 1} \right)$

[DO NOT PROVE THIS]

Prove that QS is parallel to the normal to the hyperbola at P .

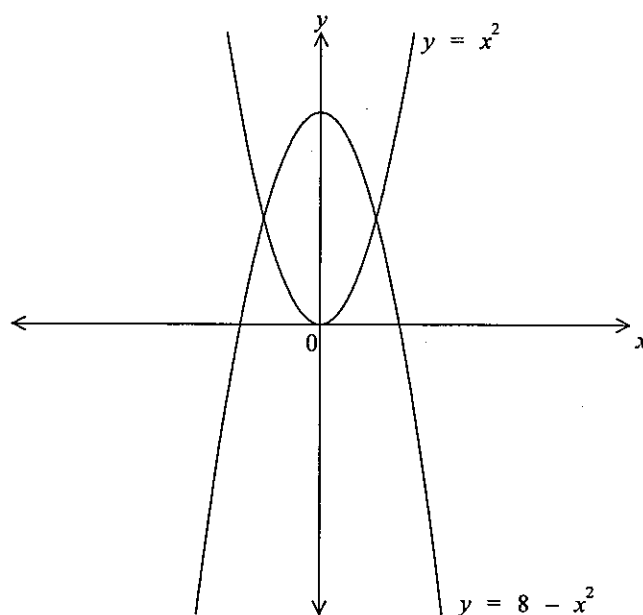
3

- (γ) Explain why RS is parallel to the tangent to the hyperbola at P .

2

Question 3 continues next page

(b)



The diagram shows the two parabolae $y = x^2$ and $y = 8 - x^2$

A solid is formed using the region enclosed between the two parabolae as its base.

Cross-sections parallel to the y axis and perpendicular to the xy plane are semi-circles where the diameters are in the base of the solid.

Prove that the volume of this solid is $\frac{256\pi}{15}$ cubic units.

5

End of Question 3

(a) A particular curve passes through the origin and its derivative is given by $\frac{dy}{dx} = \sqrt{4y^2 + 1}$

(i) Prove that $\frac{d^2y}{dx^2} = 4y$ 2

(ii) Use the table of standard integrals to find x as a function of y . 2

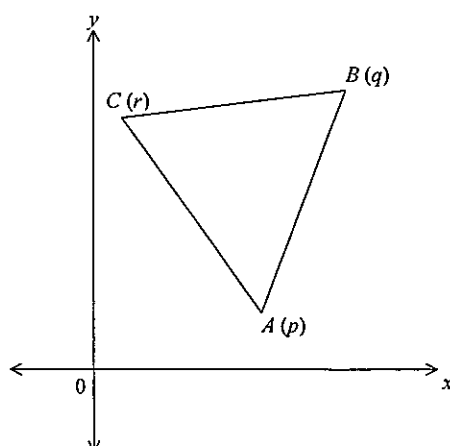
(b) (i) Sketch the region where $0 \leq y \leq x - x^2$ 1

(ii) The region in (i) is revolved about the line $x = -1$

Use the method of cylindrical shells to find the volume of the solid of revolution generated. 4

(c) (i) Express $\frac{1}{2}(1 + i\sqrt{3})$ in mod-arg form. 1

(ii)



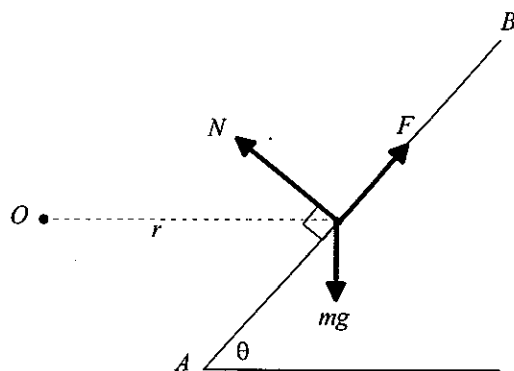
In the Argand diagram the points A, B, C represent the complex numbers p, q, r , respectively, where $r - p = \frac{1}{2}(1 + i\sqrt{3})(q - p)$

(α) Prove that $p - q = \frac{1}{2}(1 + i\sqrt{3})(r - q)$ 3

(β) Deduce that $p^2 + q^2 + r^2 = pq + qr + rp$ 2

End of Question 4

(a)



The diagram shows the forces exerted on a car of mass m travelling at speed v on a banked circular track AB of radius r . The track is banked inwards at θ to the horizontal. The road exerts the normal force N at right angles to the road and there is a frictional force F exerted up the track.

- (i) By resolving the forces in the direction BA, or otherwise, show that

$$F = mg \sin \theta - \frac{mv^2}{r} \cos \theta \quad 2$$

- (ii) Deduce that $v^2 < gr \tan \theta$ 2

- (iii) Draw a diagram showing the forces on the car if $v^2 > gr \tan \theta$ 1

- (iv) Find an expression for N not involving F . 2

- (b) u, v, w are the roots of $x^3 + Ax + B = 0$

- (i) Show that $u^2 + v^2 + w^2 = -2A$ 2

- (ii) Let $y = \frac{v}{w} + \frac{w}{v}$
Prove that $u^3 + 2Au - By = 0$ 2

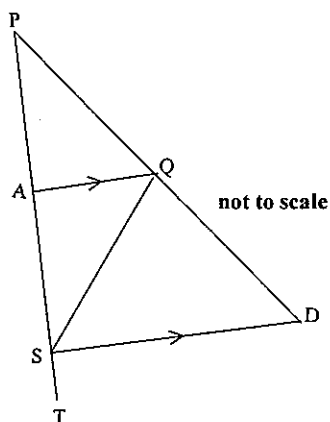
- (iii) By using another equation involving u^3 show that $u = \frac{B}{A}(y + 1)$ 2

- (iv) Show that the equation with roots $\frac{v}{w} + \frac{w}{v}, \frac{w}{u} + \frac{u}{w}$ and $\frac{u}{v} + \frac{v}{u}$ is
 $B^2(x + 1)^3 + A^3(x + 1) + A^3 = 0$ 1

- (v) Evaluate $\frac{v}{w} + \frac{w}{v} + \frac{w}{u} + \frac{u}{w} + \frac{u}{v} + \frac{v}{u}$ 1

End of Question 5

(a)



In the diagram PAST and PQD are straight lines and $AQ \parallel SD$

Further, $\frac{PS}{QS} = \frac{PD}{QD}$

(i) Explain why $QS = AS$

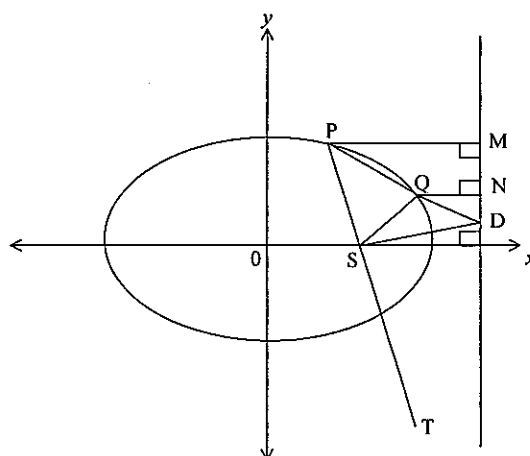
1

(ii) Deduce that $\angle DSQ = \angle DST$

2

Question 6 continues next page

(b)



The diagram shows a chord PQ of an ellipse meeting a directrix at D. S is the corresponding focus. PM and QN meet this directrix at right angles at M and N, respectively.

(i) Show that $\frac{PS}{QS} = \frac{PM}{QN}$ 1

(ii) Deduce that $\frac{PS}{QS} = \frac{PD}{QD}$ 1

(iii) Deduce that $\angle DSQ = \angle DST$ 1

(iv) Deduce that if the tangent at P meets the directrix at R then $\angle PSR = 90^\circ$ 2

(c) (i) w is a complex root of $x^3 - 1 = 0$

(α) Explain why $\overline{w}$ is the other complex root and deduce that $1 + w + \overline{w} = 0$ 2

(β) Show that $\overline{w} = w^2$ 1

(ii) $A(x)$ and $B(x)$ are two polynomials with complex coefficients such that $A(x^3) + xB(x^3) \equiv (x^2 + x + 1)Q(x)$, where $Q(x)$ is a polynomial with complex coefficients.

(α) Prove that $A(1) = 0$ and $B(1) = 0$ 3

(β) Deduce that $A(x^3) + xB(x^3)$ is divisible by $x^3 - 1$ 1

End of Question 6

(a) You are given the identity $\cos(A + B) + \cos(A - B) \equiv 2 \cos A \cos B$

(i) Evaluate $\int_0^{\frac{\pi}{4}} \cos 5x \cos 3x \, dx$ 2

(ii) Find the general solutions of the equation $\cos 5x + \cos 3x + 2\cos x = 0$ 3

(b) A particle of unit mass moves on the x axis against a resistance numerically equal to $v^2 + v^3$, where v is its velocity. Initially the particle is travelling with velocity u , where $u > 0$.

(i) Prove that when the velocity is $\frac{u}{2}$ the distance X travelled by the particle is given by $X = \ln\left(\frac{2 + u}{1 + u}\right)$ 4

(ii) Prove that if T is the time taken to travel the distance X then $u(T + X) = 1$ 4

(iii) Thomas examined the motion of the particle more thoroughly. Thomas alleged that if the particle started at the origin then the velocity v , displacement x and time t were related by the equation

$$v = \frac{u}{ux + ut + 1} \quad 2$$

By finding a suitable derivative, show that Thomas is correct.

End of Question 7

(a) Let $u_n = \int_0^1 x^{2007} (1-x)^n dx$, $n = 0, 1, 2, \dots$

(i) By considering $u_n - u_{n-1}$ show that $u_n < u_{n-1}$ 2

(ii) Use integration by parts to show that $u_n = \frac{n}{2008+n} u_{n-1}$, $n \geq 1$ 3

(iii) Deduce that $u_n = \frac{2007! n!}{(2008+n)!}$ 2

○ (b) An extraordinary identity, due to the Swiss mathematician Leonard Euler (1707-83), states

$$e^{i\theta} = \cos \theta + i \sin \theta \text{ for all real values of } \theta$$

(i) Show that $\cos \theta = \frac{1}{2}(e^{i\theta} + e^{-i\theta})$ and $\sin \theta = \frac{1}{2i}(e^{i\theta} - e^{-i\theta})$ 2

(ii) Deduce that

$$1 + 2 \cos \theta + 2 \cos 2\theta + \dots + 2 \cos n\theta = \frac{\sin\left(n + \frac{1}{2}\right)\theta}{\sin \frac{1}{2}\theta}, \quad n \geq 1 \quad 3$$

(iii) Hence or otherwise find $\lim_{\theta \rightarrow 0} \frac{\sin\left(n + \frac{1}{2}\right)\theta}{\sin \frac{1}{2}\theta}$ 1

○ (iv) Use (ii) to show that

$$1 + (2 \cos \theta)^2 + (2 \cos 2\theta)^2 + \dots + (2 \cos n\theta)^2 = 2n + \frac{\sin(2n+1)\theta}{\sin \theta}, \quad n \geq 1 \quad 2$$

End of Examination

Standard Integrals

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

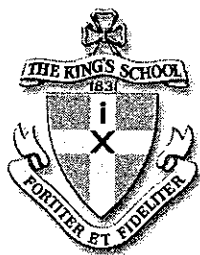
$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

Note: $\ln x = \log_e x, \quad x > 0$



THE KING'S SCHOOL

2007
Higher School Certificate
Trial Examination

Mathematics Extension 2

Question	(Marks)	Complex Numbers	Functions	Integration	Conics	Mechanics
1	(15)			15		
2	(15)	(a), (b), (c) 9	(d) 6			
3	(15)			(b) 5	(a) 10	
4	(15)	(c) 5		(a), (b) 10		
5	(15)		(b) 8			(a) 7
6	(15)		(c) 7		(a), (b) 8	
7	(15)			(a) 5		(b) 10
8	(15)	(b) 8		(a) 7		
Total	(120)	22	21	42	18	17

Question 1

(a) $x = \tan \frac{x}{2}$

$x=0, t=0$

$\frac{dt}{dx} = \frac{1}{2} \sec^2 \frac{x}{2} = \frac{1}{2} (1+t^2)$

$x = \frac{\pi}{2}, t=1$

$$\therefore I = \int_0^1 \frac{2 dt}{3(1+t^2) + 1-t^2}$$

$$= \int_0^1 \frac{dt}{2+t^2} = \frac{1}{\sqrt{2}} \left[\tan^{-1} \frac{t}{\sqrt{2}} \right]_0^1 = \frac{1}{\sqrt{2}} \tan^{-1} \frac{1}{\sqrt{2}}$$

(b) (i) put $u = 1-x^2$; $x=0, u=1$

$\frac{du}{dx} = -2x$

$x = \frac{\sqrt{3}}{2}, u = \frac{1}{4}$

$$\therefore I = -\frac{1}{2} \int_1^{\frac{1}{4}} \frac{1}{\sqrt{u}} du = \frac{1}{2} \cdot 2 \left[\sqrt{u} \right]_{\frac{1}{4}}^1 = \frac{1}{2}$$

(ii) $u = \cos^{-1} x$, $\frac{du}{dx} = 1$

$\frac{du}{dx} = -\frac{1}{\sqrt{1-x^2}}$, $v=x$

$$\therefore I = \left[x \cos^{-1} x \right]_0^{\frac{\sqrt{3}}{2}} + \int_0^{\frac{\sqrt{3}}{2}} \frac{x}{\sqrt{1-x^2}} dx$$

$$= \frac{\pi \sqrt{3}}{12} + \frac{1}{2}$$

(c) (i) Equating constants, $-2A - 2 = -12$
 $\therefore A = 5$

$$\begin{aligned}
 \text{(ii) } I &= \int_0^1 \frac{2x+5}{x^2+1} - \frac{2}{x-2} dx \\
 &= \int_0^1 \frac{2x}{x^2+1} + \frac{5}{x^2+1} - \frac{2}{x-2} dx \\
 &= \left[\ln(x^2+1) + 5 \tan^{-1}x - 2 \ln|x-2| \right]_0^1 \\
 &= \ln 2 + \frac{5\pi}{4} - 0 - (0 + 0 - 2 \ln 2) \\
 &= 3 \ln 2 + \frac{5\pi}{4}
 \end{aligned}$$

$$\begin{aligned}
 \text{(d) } I &= \int \tan x \tan^2 x dx \\
 &= \int \tan x (\sec^2 x - 1) dx \\
 &= \int \tan x \sec^2 x - \frac{\sin x}{\cos x} dx \\
 &= \frac{\tan^2 x}{2} + \ln \cos x (+ c)
 \end{aligned}$$

Question 2

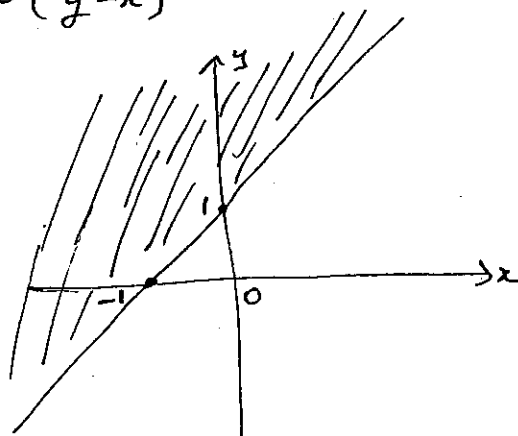
(a) (i) $\left| \frac{z}{w} \right| = \frac{\sqrt{a^2+1}}{\sqrt{1+a^2}} = 1$

(ii) $zw = (a+i)(1+ai) = a - a + i(a^2+1)$
 $= (a^2+1)i$

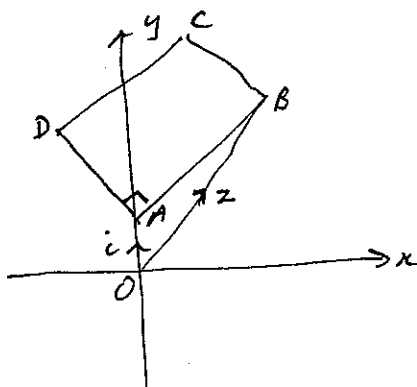
$\therefore \arg zw = \frac{\pi}{2}$

(b) $(1-i)(x+iy) = x+y+i(y-x)$

$\therefore y-x \geq 1$



(c) (i)



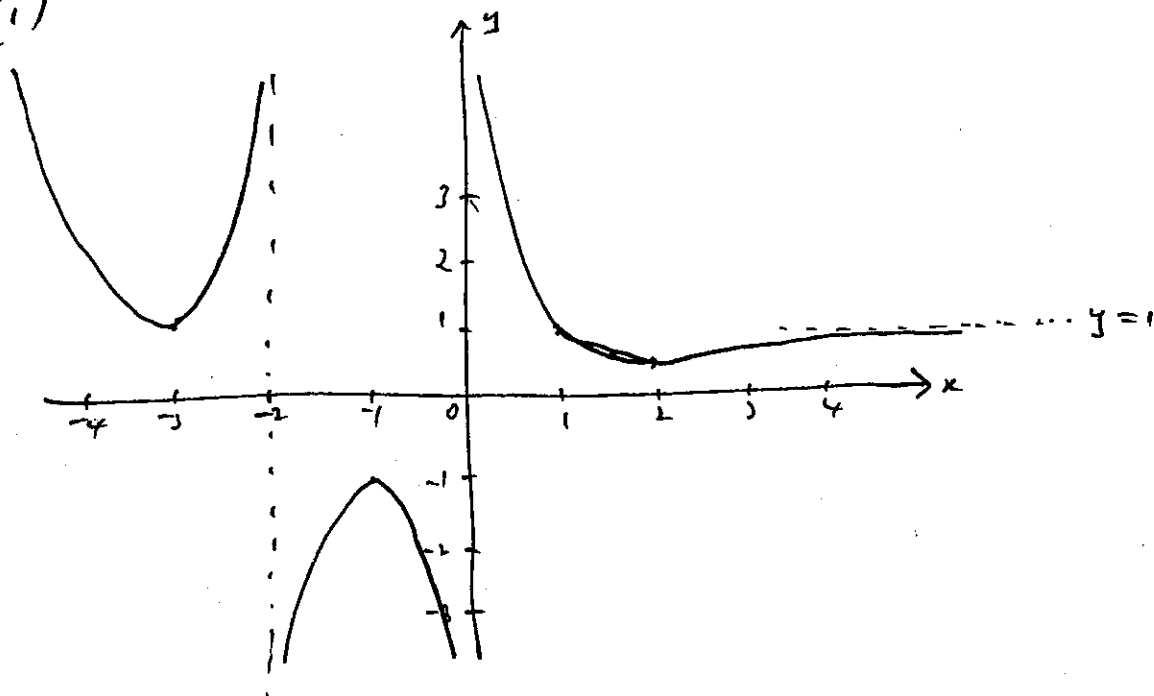
$\vec{AB} = z - i$

$\therefore \vec{AD} = i \vec{AB} = 1 + iz$

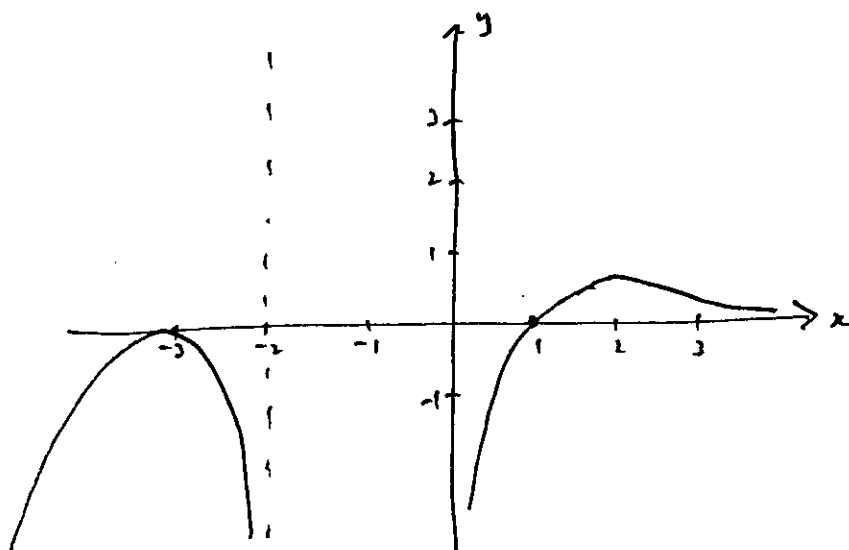
(ii) From (i), $\vec{BC} = 1 + iz$

$\therefore \vec{OC} = \vec{OB} + \vec{BC} = z + 1 + iz$
 $= 1 + (1+i)z$

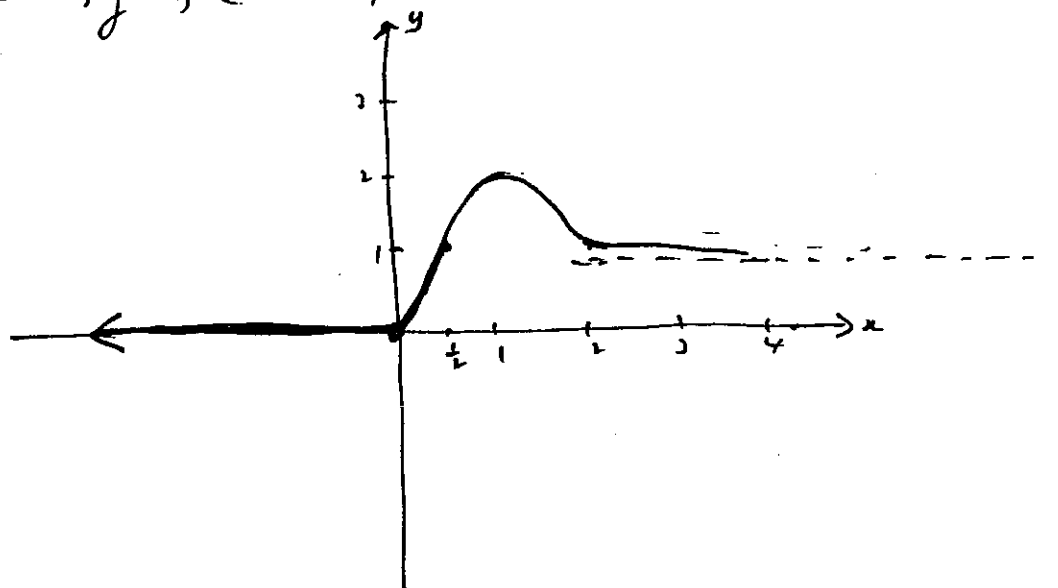
(d) (i)



(ii)



(iii) If $x \geq 0$, $y = f(x+x) = f(2x)$
 If $x \leq 0$, $y = f(x-x) = f(0) = 0$

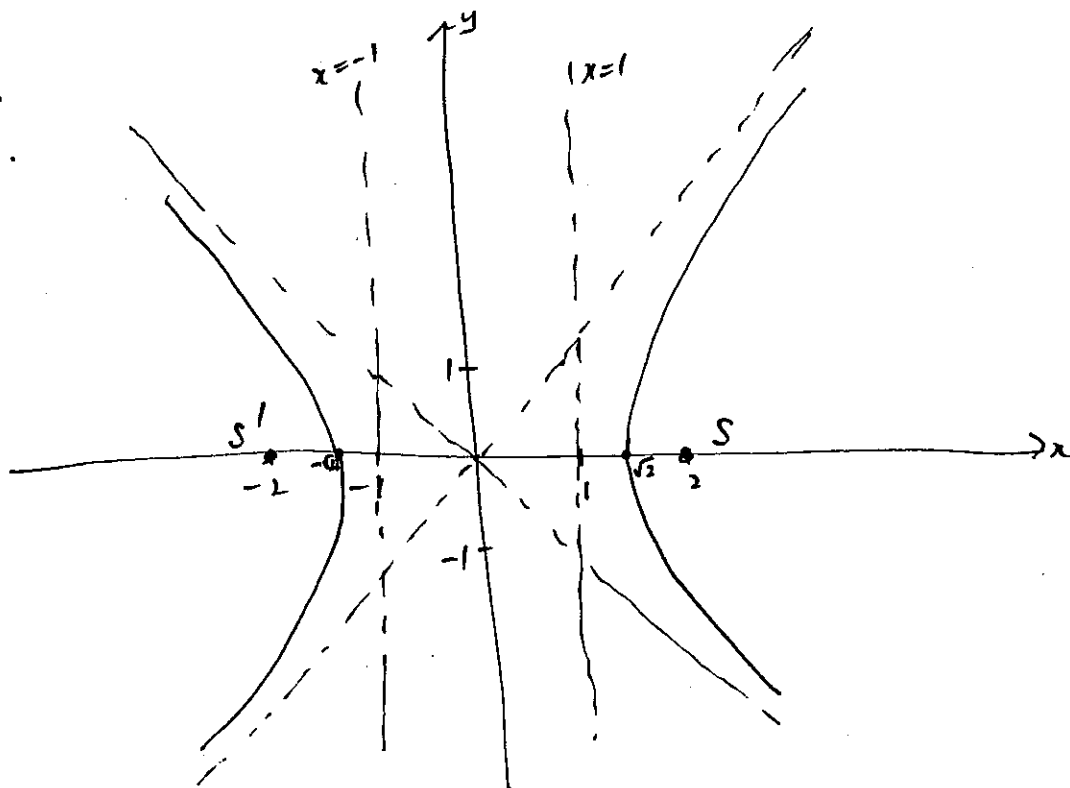


Question 3

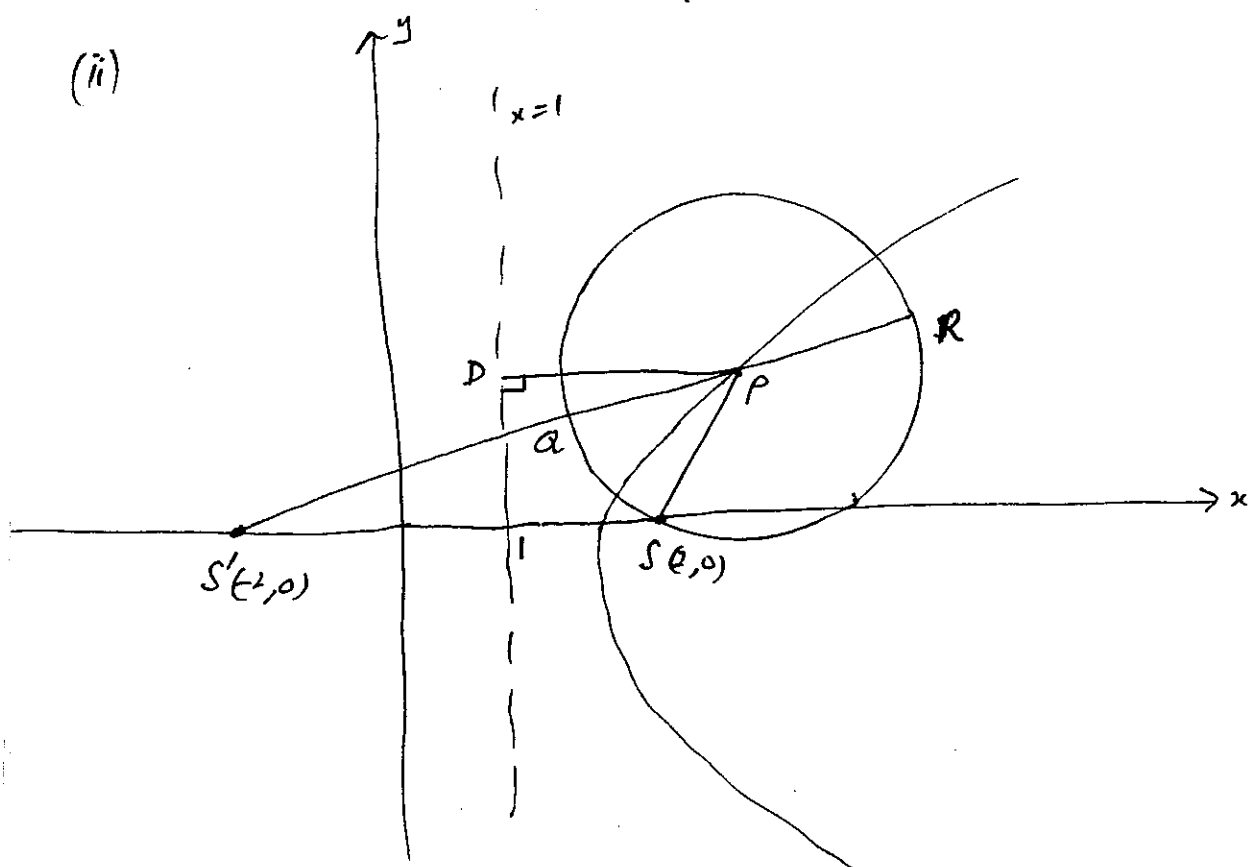
(a) (i) $c^2 = 2+2=4 \therefore c=2$ ie. $S=(2,0)$

$e=\sqrt{2} \Rightarrow$ directrices are $x = \pm \frac{\sqrt{2}}{\sqrt{2}} = \pm 1$

$y=0 \Rightarrow x = \pm \sqrt{2}$



(ii)



$$(2) \quad PS = e \cdot PD = \sqrt{2} (\sqrt{2} \sec \theta - 1) = 2 \sec \theta - \sqrt{2}$$

$$\begin{aligned} (\beta) \quad \text{gradient } QS &= \frac{2\sqrt{2} \tan \theta}{(\sqrt{2} \sec \theta + 1) \left[\frac{2}{\sqrt{2} \sec \theta + 1} - 2 \right]} \\ &= \frac{\sqrt{2} \tan \theta}{1 - (\sqrt{2} \sec \theta + 1)} = -\frac{\tan \theta}{\sec \theta} \end{aligned}$$

Next, $2x - 2y \frac{dy}{dx} = 0$ on hyperbola

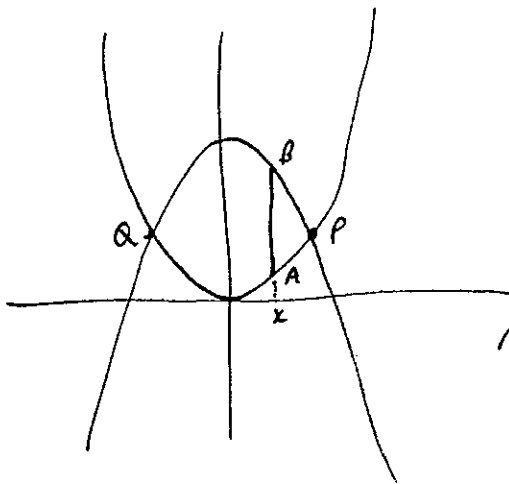
$$\therefore \frac{dy}{dx} = \frac{x}{y} = \frac{\sec \theta}{\tan \theta} \quad \text{at } P$$

$$\therefore \text{gradient normal} = -\frac{\tan \theta}{\sec \theta} \Rightarrow QS \parallel \text{normal}$$

(f) Now, $\angle QSR = 90^\circ$, angle in semi-circle

$$\therefore RS \perp QS \Rightarrow RS \parallel \text{tangent at } P \text{ from } (P)$$

(b)



$$\begin{aligned} \text{at } P, Q, \quad x^2 &= 8 - x^2 \\ x^2 &= 4 \Rightarrow x = 2, -2 \end{aligned}$$

$$AB = 8 - x^2 - x^2 = 8 - 2x^2$$

$$\begin{aligned} \text{Area of semi-circle on } AB \text{ as diameter } &= \frac{1}{2} \pi (4 - x^2)^2 \end{aligned}$$

$$\therefore V = 2 \int_0^2 \frac{1}{2} \pi (4 - x^2)^2 dx$$

$$= \pi \int_0^2 (16 - 8x^2 + x^4) dx$$

$$= \pi \left[16x - \frac{8x^3}{3} + \frac{x^5}{5} \right]_0^2 = \pi \left(32 - \frac{64}{3} + \frac{32}{5} \right) = \frac{256\pi}{15}$$

Question 4

$$(a) (i) \quad \frac{dy^2}{dx^2} = \frac{1}{2} (4y^2 + 1)^{-\frac{1}{2}} \cdot 8y \cdot \frac{dy}{dx}$$
$$= 4y \cdot \frac{dx}{dy} \cdot \frac{dy}{dx} = 4y$$

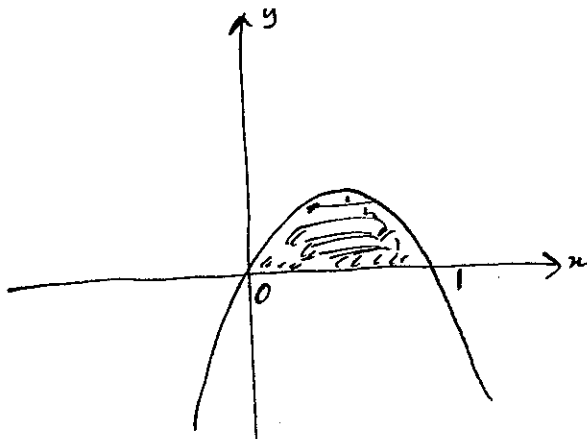
$$(ii) \quad \frac{dx}{dy} = \frac{1}{\sqrt{4y^2 + 1}} = \frac{1}{\sqrt{(2y)^2 + 1}}$$

$$\therefore x = \frac{1}{2} \ln(2y + \sqrt{4y^2 + 1}) + c$$

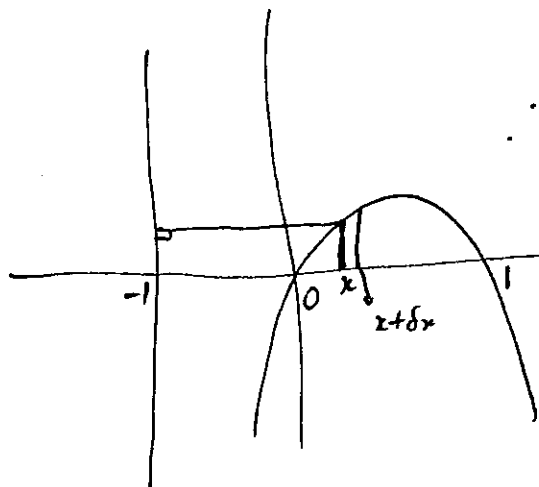
$$\therefore 0 = \frac{1}{2} \ln 1 + c, \quad c = 0$$

$$\therefore x = \frac{1}{2} \ln(2y + \sqrt{4y^2 + 1})$$

(b) (i)



(ii)



$$\therefore \delta V \approx \pi [(x + \delta x + 1)^2 - (x + 1)^2] y$$

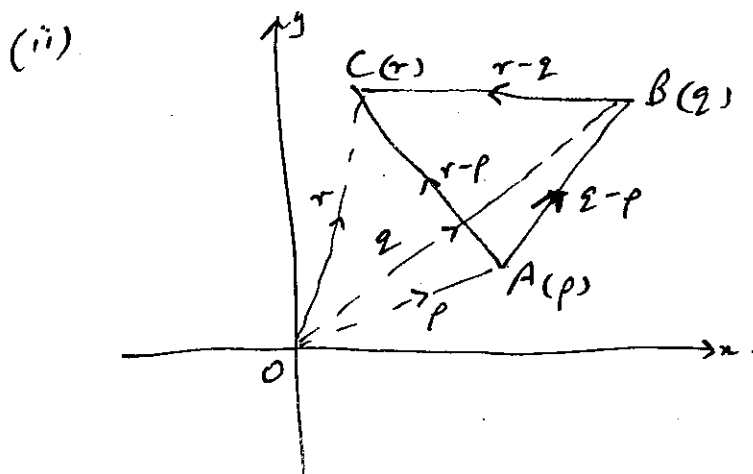
$$\approx \pi (2(x + 1) \delta x) y$$

$$\Rightarrow V = 2\pi \int_0^1 (x + 1)(x - x^2) dx$$

$$= 2\pi \int_0^1 x - x^3 dx$$

$$= 2\pi \left[\frac{x^2}{2} - \frac{x^4}{4} \right]_0^1 = \frac{\pi}{2}$$

$$(c) \quad (i) \quad \cos \frac{\pi}{3} + i \sin \frac{\pi}{3}$$



$$(d) \quad \vec{AC} = r - p, \quad \vec{AB} = q - p$$

$$\therefore \vec{AC} = \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right) \vec{AB}$$

$$\Rightarrow \angle CAB = 60^\circ \quad \& \quad |\vec{AC}| = |\vec{AB}|$$

$$\therefore \angle ACB = \angle ABC = 60^\circ$$

$\therefore \triangle ABC$ is equilateral

$$\therefore \vec{BA} = \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right) \vec{BC}$$

$$\Rightarrow p - q = \frac{1}{2} (1 + i\sqrt{3}) (r - q)$$

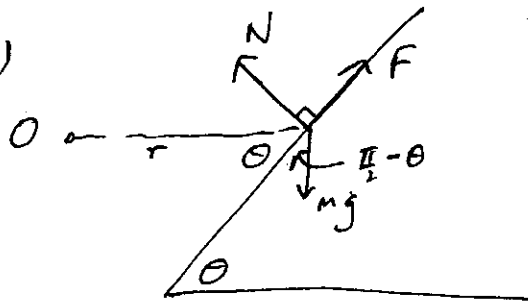
$$(e) \quad \therefore \frac{r - p}{p - q} = \frac{q - p}{r - q} \quad \text{from data \& (d)}$$

$$\therefore r^2 - pr - qr + pq = -(p^2 - 2pq + q^2)$$

$$\text{i.e.} \quad p^2 + q^2 + r^2 = pq + qr + rp$$

Question 5

(a) (i)



Resolving in direction BA we have

$$\frac{mv^2}{r} \cos \theta = mg \cos(\frac{\pi}{2} - \theta) - F$$

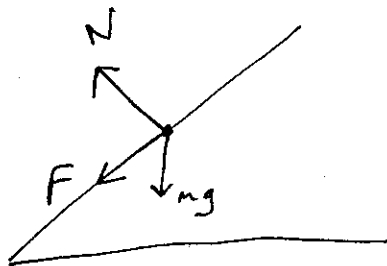
$$\Rightarrow F = mg \sin \theta - m \frac{v^2}{r} \cos \theta$$

(ii) Since $F > 0$, $g \sin \theta - \frac{v^2}{r} \cos \theta > 0$

$$\text{i.e. } g \tan \theta - \frac{v^2}{r} > 0$$

$$\Rightarrow v^2 < rg \tan \theta$$

(iii)



(iv) Resolving in the direction of N,

$$m \frac{v^2}{r} \cos(\frac{\pi}{2} - \theta) = N - mg \cos \theta$$

$$\Rightarrow N = mg \cos \theta + m \frac{v^2}{r} \sin \theta$$

$$(b) (i) u^2 + v^2 + w^2 = (u+v+w)^2 - 2(uv + vw + wu) \\ = 0^2 - 2A = -2A$$

$$(ii) y = \frac{v^2 + w^2}{vw} \quad \text{where } u^2 + v^2 + w^2 = -2A \\ \text{and } uvw = -B$$

$$\therefore y = \frac{-2A - u^2}{-\frac{B}{u}} = \frac{2Au + u^3}{B}$$

$$\text{or } u^3 + 2Au - By = 0$$

$$(iii) \text{ Since } u \text{ is a root then} \\ u^3 + Au + B = 0$$

$$\therefore \text{ from (ii), } Au - By - B = 0$$

$$\therefore u = \frac{B}{A}(y+1)$$

(iv) From (iii) the equation is

$$\left(\frac{B}{A}(y+1)\right)^3 + A \frac{B}{A}(y+1) + B = 0$$

$$\text{i.e. } B^2(y+1)^3 + A^3(y+1) + A^3 = 0$$

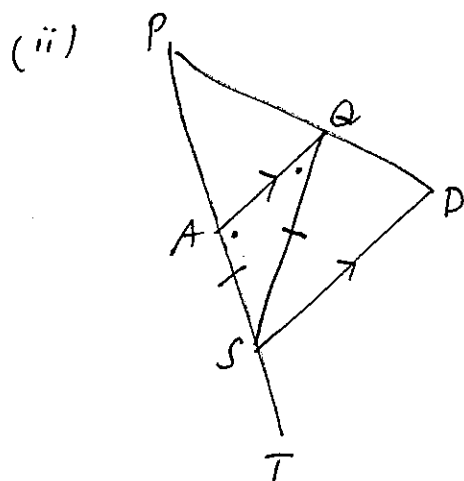
$$\text{or } B^2(x+1)^3 + A^3(x+1) + A^3 = 0$$

(v) i.e. the sum of the roots of (iv)

$$= -\frac{3B^2}{B^2} = -3$$

Question 6

(a) (i) $\frac{PD}{QD} = \frac{PS}{AS}$, ratio intercept theorem in $\parallel$ lines
 $= \frac{PS}{QS}$, data $\therefore QS = AS$



From (i), $\angle SQA = \angle SAQ$, base angles
 isos Δ .

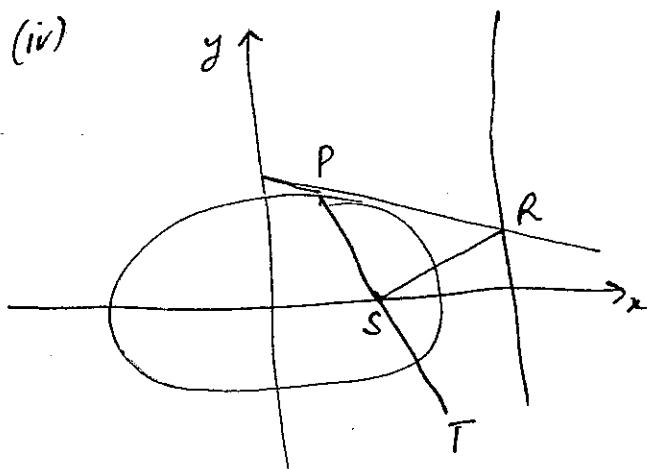
But $\angle SQA = \angle DSQ$, alt $\angle$ s in $\parallel$ lines
 $\& \angle SAQ = \angle DST$, corr. $\angle$ s in $\parallel$ lines
 $\therefore \angle DSQ = \angle DST$

(b) (i) $\frac{PS}{QS} = \frac{ePM}{eQN} = \frac{PM}{QN}$, (focus-directrix defn of ellipse)

(ii) $\frac{PM}{QN} = \frac{PD}{QD}$ since $\Delta PMD \parallel \Delta QND$

$\therefore$ from (i), $\frac{PS}{QS} = \frac{PD}{QD}$

(iii) $\therefore \angle DSQ = \angle DST$ from (a), independent of A



as $A \rightarrow P$, $D \rightarrow R$

$\Rightarrow$ from (iii),

$\angle RSP = \angle RST$

But $\angle PST = 180^\circ$

$\therefore \angle PSR = 90^\circ$

(c) (i) (2) Since the coefficients of $x^3 - 1 = 0$ are real
then the complex roots occur in conjugate pairs.

$\therefore \bar{\omega}$ is the other complex root

$\therefore 1 + \omega + \bar{\omega} = 0$ since sum of roots is 0

$$(p) \quad x^3 - 1 = (x-1)(x^2 + x + 1)$$

$$\Rightarrow 1 + \omega + \omega^2 = 0 \Rightarrow \bar{\omega} = \omega^2, \text{ from (2)}$$

[Alternatives abound]

$$(ii) \quad (2), \therefore A(\omega^3) + \omega B(\omega^3) = 0$$

$$\therefore A(1) + \omega B(1) = 0 \text{ since } 1 + \omega + \omega^2 = 0$$

$$\text{Also, } A(\bar{\omega}^3) + \bar{\omega} B(\bar{\omega}^3) = 0 \text{ since } 1 + \bar{\omega} + \bar{\omega}^2 = 0$$

$$\text{i.e. } A(1) + \bar{\omega} B(1) = 0$$

$$\therefore (\omega - \bar{\omega}) B(1) = 0 \Rightarrow B(1) = 0 \text{ since } \omega \neq \bar{\omega}$$

$$\therefore A(1) = 0$$

$$(p) \quad \text{From (2), } 1 \text{ is a root of } A(x^3) + x B(x^3) = 0$$

$$\Rightarrow A(x^3) + x B(x^3) \equiv (x^2 + x + 1)(x-1) R(x)$$
$$= (x^3 - 1) R(x)$$

i.e. is divisible by $x^3 - 1$

Question 7

$$(a) (i) \therefore I = \frac{1}{2} \int_0^{\frac{\pi}{4}} \cos 8x + \cos 2x \, dx$$

$$= \frac{1}{2} \left[\frac{\sin 8x}{8} + \frac{\sin 2x}{2} \right]_0^{\frac{\pi}{4}}$$

$$= \frac{1}{2} \left(0 + \frac{1}{2} - 0 \right) = \frac{1}{4}$$

$$(ii) \text{ For } \cos 5x + \cos 3x \text{ put } \begin{aligned} A+B &= 5x \\ A-B &= 3x \end{aligned}$$

$$\Rightarrow A=4x, B=x$$

$$\therefore \text{ we have } 2 \cos 4x \cos x + 2 \cos x = 0$$

$$\text{or } \cos x (\cos 4x + 1) = 0$$

$$\therefore \cos x = 0 \text{ or } \cos 4x = -1$$

$$\therefore x = 2n\pi \pm \cos^{-1} 0 \text{ or } 4x = 2n\pi \pm \cos^{-1}(-1)$$

$$\text{i.e. } x = 2n\pi \pm \frac{\pi}{2} \text{ or } x = \frac{n\pi}{2} \pm \frac{\pi}{4}, n \text{ an integer}$$

$$(b) (i) \quad v \frac{dv}{dx} = -(v^2 + v^3)$$

$$\therefore \frac{dv}{dx} = -(v + v^2)$$

$$\Rightarrow \frac{dx}{dv} = -\frac{1}{v(1+v)} = \frac{1}{1+v} - \frac{1}{v}$$

$$\therefore X = \int_u^{\frac{u}{2}} \left(\frac{1}{1+v} - \frac{1}{v} \right) dv$$

$$= \left[\ln(1+v) - \ln v \right]_u^{\frac{u}{2}}$$

$$= \left[\ln \left(\frac{1+v}{v} \right) \right]_u^{\frac{u}{2}} = \ln \left(\frac{1 + \frac{u}{2}}{\frac{u}{2}} \right) - \ln \left(\frac{1+u}{u} \right)$$

$$= \ln \left(\frac{2+u}{u} \cdot \frac{u}{1+u} \right)$$

$$= \ln \left(\frac{2+u}{1+u} \right)$$

$$(ii) \quad \frac{dv}{dt} = -(v^2 + v^3)$$

$$\Rightarrow \frac{dt}{dv} = -\frac{1}{v^2(1+v)} \equiv \frac{Av+B}{v^2} + \frac{C}{1+v}$$

$$\Rightarrow (1+v)(Av+B) + Cv^2 \equiv -1$$

$$\therefore B = -1, C = -1, A = 1$$

$$\therefore T = \int_u^{\frac{u}{2}} \frac{v-1}{v^2} - \frac{1}{1+v} dv$$

$$= \int_u^{\frac{u}{2}} \frac{1}{v} - \frac{1}{1+v} - \frac{1}{v^2} dv$$

$$= -X - \int_u^{\frac{u}{2}} \frac{1}{v^2} dv \quad \text{from (i)}$$

$$= -X + \left[\frac{1}{v} \right]_u^{\frac{u}{2}}$$

$$\text{i.e. } T + X = \frac{2}{u} - \frac{1}{u} = \frac{1}{u}$$

$$\therefore u(T + X) = 1$$

$$(iii) \quad v = u(ux + ut + 1)^{-1}$$

$$\therefore \frac{dv}{dt} = -u(ux + ut + 1)^{-2} \left(u \frac{dx}{dt} + u \right)$$

$$= \frac{-u(uv + u)}{(ux + ut + 1)^2}$$

$$\Rightarrow \frac{dv}{dt} = - \left(\frac{u}{ux + ut + 1} \right)^2 (v + 1)$$

$$= -v^2(v + 1)$$

$$= -(v^2 + v^3)$$

= Thomas is correct

Question 8

$$(a) (i) u_n - u_{n-1} = \int_0^1 x^{2007} [(1-x)^n - (1-x)^{n-1}] dx$$

$$= \int_0^1 x^{2007} (1-x)^{n-1} [1-x-1] dx$$

$$= - \int_0^1 x^{2007} (1-x)^{n-1} dx$$

$$< 0 \text{ since } x^{2007} (1-x)^{n-1} \geq 0 \text{ for } 0 \leq x \leq 1$$

$$\text{i.e. } u_n < u_{n-1}$$

$$(ii) \text{ Put } u = (1-x)^n, \quad \frac{du}{dx} = -n(1-x)^{n-1}$$

$$\text{Then } \frac{du}{dx} = -n(1-x)^{n-1}, \quad v = \frac{x^{2008}}{2008}$$

$$\therefore u_n = \left[\frac{x^{2008} (1-x)^n}{2008} \right]_0^1 + \frac{n}{2008} \int_0^1 x^{2007} (1-x)^{n-1} dx$$

$$= 0 + \frac{n}{2008} (u_{n-1} - u_n) \text{ from (i)}$$

$$\therefore u_n \left(1 + \frac{n}{2008} \right) = \frac{n}{2008} u_{n-1}$$

$$\text{i.e. } u_n = \frac{n}{2008+n} u_{n-1}$$

$$(iii) \text{ From (ii), } u_n = \frac{n}{2008+n} \cdot \frac{n-1}{2007+n} \cdot \frac{n-2}{2006+n} \cdots \frac{1}{2009} u_0$$

$$\text{where } u_0 = \int_0^1 x^{2007} dx = \frac{1}{2008}$$

$$\therefore u_n = \frac{n}{2008+n} \cdot \frac{n-1}{2007+n} \cdots \frac{2}{2010} \cdot \frac{1}{2009} \cdot \frac{1}{2008} \cdot \frac{2007 \cdot 2006 \cdots 2 \cdot 1}{2007 \cdot 2006 \cdots 2 \cdot 1}$$

$$= \frac{n! \cdot 2007!}{(2008+n)!}$$

$$(h) (i) e^{i\theta} = \cos \theta + i \sin \theta$$

$$\therefore e^{-i\theta} = \cos(-\theta) + i \sin(-\theta) = \cos \theta - i \sin \theta$$

$$\therefore 2 \cos \theta = e^{i\theta} + e^{-i\theta} \quad \text{and} \quad 2i \sin \theta = e^{i\theta} - e^{-i\theta}$$

$$\therefore \cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2} \quad \text{and} \quad \sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}$$

(ii) From (i),

$$L.S = 1 + (e^{i\theta} + e^{-i\theta}) + (e^{2i\theta} + e^{-2i\theta}) + \dots + (e^{ni\theta} + e^{-ni\theta})$$

$$= e^{-in\theta} + e^{-i(n-1)\theta} + \dots + e^{-i\theta} + 1 + e^{i\theta} + \dots + e^{in\theta}$$

$$= \frac{e^{-in\theta} \left((e^{i\theta})^{2n+1} - 1 \right)}{e^{i\theta} - 1} \quad \text{using the geometric series}$$

$$= \frac{e^{i(n+1)\theta} - e^{-in\theta}}{e^{i\theta} - 1}$$

$$= \frac{e^{i(n+\frac{1}{2})\theta} - e^{-i(n+\frac{1}{2})\theta}}{e^{\frac{i\theta}{2}} - e^{-\frac{i\theta}{2}}}$$

$$= \frac{2i \sin(n+\frac{1}{2})\theta}{2i \sin \frac{1}{2}\theta} \quad \text{from (i)}$$

$$= \frac{\sin(n+\frac{1}{2})\theta}{\sin \frac{1}{2}\theta}$$

$$\begin{aligned} \text{(iii) Limit} &= 1 + 2 + 2 + \dots + 2, \quad n \text{ 2's} \\ &= 2n + 1 \end{aligned}$$

$$\text{(iv) } LS = 1 + 2(2\cos^2\theta) + 2(2\cos^2 2\theta) + \dots + 2(2\cos^2 n\theta)$$

$$= 1 + 2(1 + \cos 2\theta) + 2(1 + \cos 4\theta) + \dots + 2(1 + \cos 2n\theta)$$

$$= 2n + (1 + 2\cos 2\theta + 2\cos 4\theta + \dots + 2\cos 2n\theta)$$

$$= 2n + \frac{\sin(n + \frac{1}{2})2\theta}{\sin(\frac{1}{2} \cdot 2\theta)} \quad \text{from (ii)}$$

$$= 2n + \frac{\sin(2n+1)\theta}{\sin\theta}$$